Chapter 6: Options Programming

6.1 Introduction

In addition to the standard functions included with your machine, you may also have optional equipment with special programming considerations. This section tells you how to program these options.

You can contact your HFO to purchase most of these options, if your machine did not come equipped with them.

6.2 Automatic Tool Presetter (ATP)

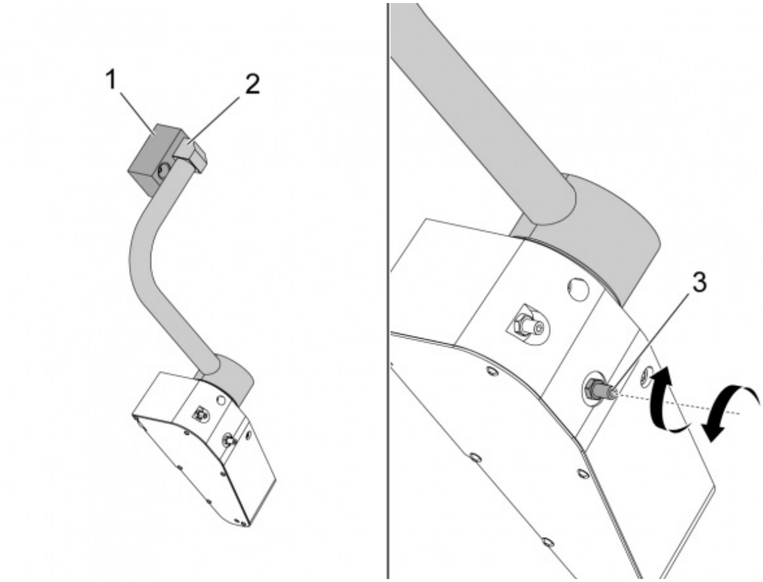
The Automatic Tool Presetter increases part accuracy and setup consistency, while reducing setup times by up to 50%. The system features easy-to-use automatic and manual modes of operation, with a user-friendly interface for quick, conversational-style programming.

- Automatic, manual, and tool-breakage-detection operations
- Increases tool setting accuracy and consistency
- Conversational-style templates for easy tool-setting operations
- No macro programming required
- Outputs G-code to MDI, where it may be edited, or transferred into a program

6.2.1 Automatic Tool Presetter (ATP) - Alignment

This procedure tells you how to align the automatic tool presetter.

1.

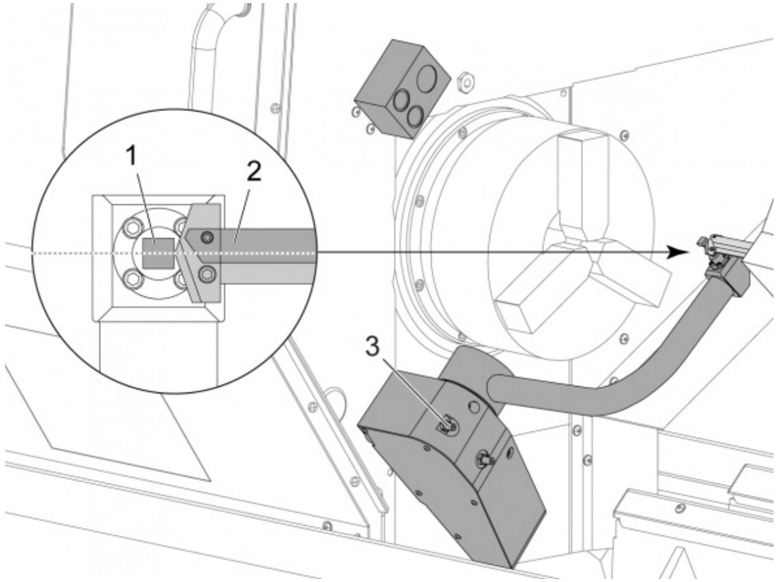


Operate this code in MDI mode for 3 minutes:

```
M104; (Tool Presetter Down)

G04 P4.;

M105; (Tool Presetter Up)

G04 P4.;

M99;
```

If the ATP arm [2] does not align with home block [1], use the 3/8-24" setscrew [3] to move it toward or away from the home block.Make sure to tighten the locknut to the adjusted position.

2.



Operate this code in MDI mode: M104. This lowers the ATP arm.

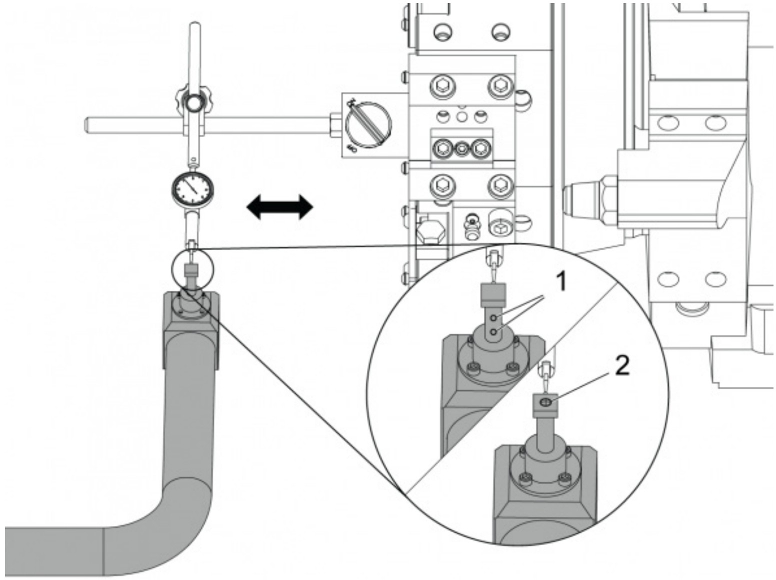
Install a turning-stick tool into the first pocket of the turret.

Jog the X and Z axes so that the tip of a turning-stick tool [2] is near the probe stylus [1].

If the tool does not align with the center of the stylus, turn the top 3/8-24" x 2" setscrew [3] to move the stylus up or down.

Make sure to tighten the locknut to the adjusted position.

3.



Attach the magnetic base of a dial indicator to the turret.

Move the indicator across the probe stylus.

The probe stylus must be parallel to the Z Axis. The error must be less than 0.0004" (0.01 mm).

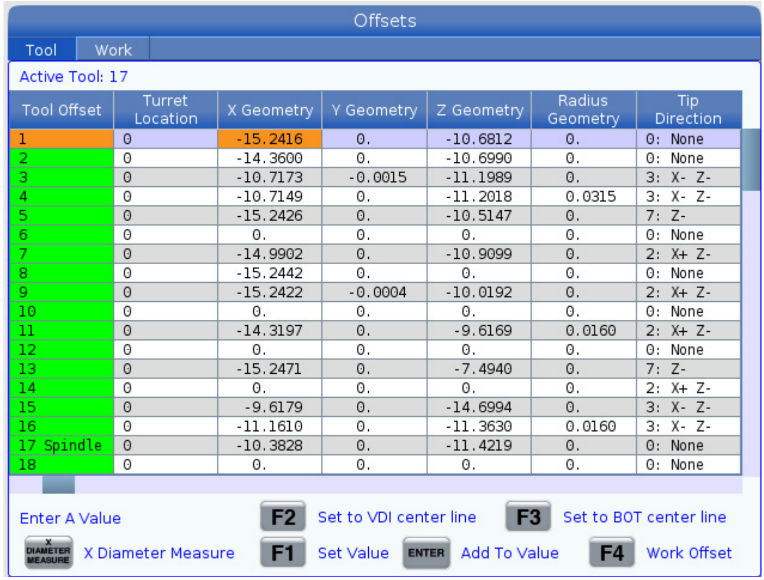
If necessary, loosen the probe stylus screws [1] [2] and adjust the position.

NOTE: *There are two types of stylus used with this ATP, one with two adjustment screws [1] and another with a single adjustment screw [2].*

6.2.2 Automatic Tool Presetter (ATP) - Test

This procedure will show you how to test the automatic tool presetter.

1.



Offsets

Tool | Work

Active Tool: 17

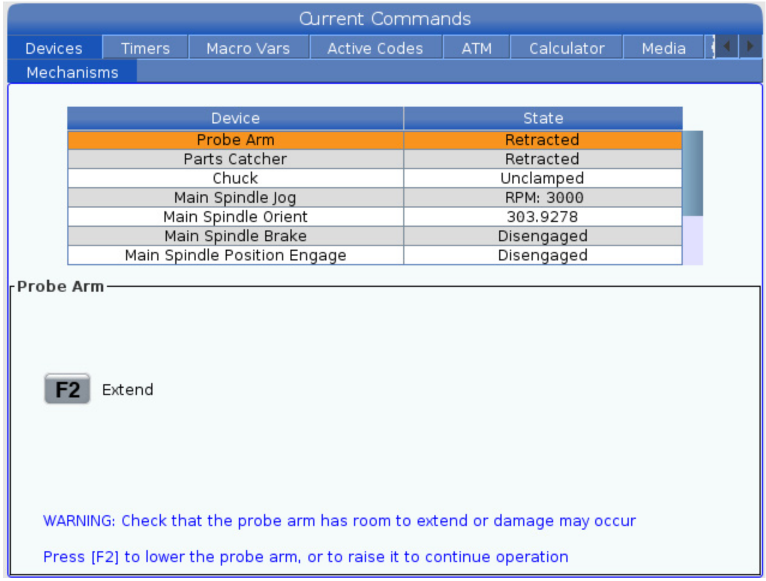
Tool Offset	Turret Location	X Geometry	Y Geometry	Z Geometry	Radius Geometry	Tip Direction
1	0	-15.2416	0.	-10.6812	0.	0: None
2	0	-14.3600	0.	-10.6990	0.	0: None
3	0	-10.7173	-0.0015	-11.1989	0.	3: X- Z-
4	0	-10.7149	0.	-11.2018	0.0315	3: X- Z-
5	0	-15.2426	0.	-10.5147	0.	7: Z-
6	0	0.	0.	0.	0.	0: None
7	0	-14.9902	0.	-10.9099	0.	2: X+ Z-
8	0	-15.2442	0.	0.	0.	0: None
9	0	-15.2422	-0.0004	-10.0192	0.	2: X+ Z-
10	0	0.	0.	0.	0.	0: None
11	0	-14.3197	0.	-9.6169	0.0160	2: X+ Z-
12	0	0.	0.	0.	0.	0: None
13	0	-15.2471	0.	-7.4940	0.	7: Z-
14	0	0.	0.	0.	0.	2: X+ Z-
15	0	-9.6179	0.	-14.6994	0.	3: X- Z-
16	0	-11.1610	0.	-11.3630	0.0160	3: X- Z-
17 Spindle	0	-10.3828	0.	-11.4219	0.	0: None
18	0	0.	0.	0.	0.	0: None

Push **[OFFSET]** until *"TOOL GEOMETRY"* is selected.

Record the value in the OFFSET

CAUTION: *Be sure to accurately record this value.*

2.



Make sure that the ATP arm does not hit parts of the machine.

Push **[CURRENT COMMANDS]**.

Select the Devices tab.

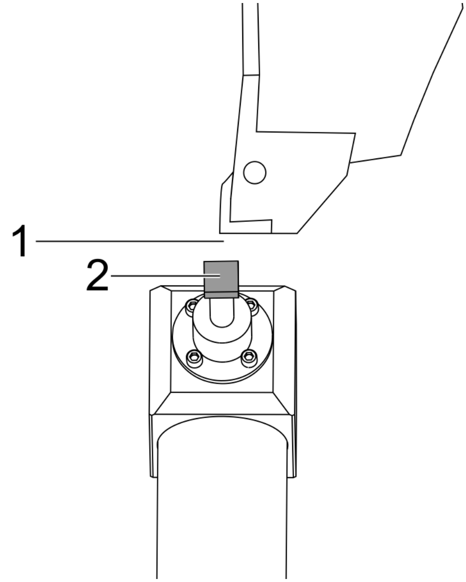
Select the Mechanisms tab.

Highlight Probe Arm.

Push **[F2]** to lift the ATP arm.

Push **[F2]** to lower the ATP arm.

3.



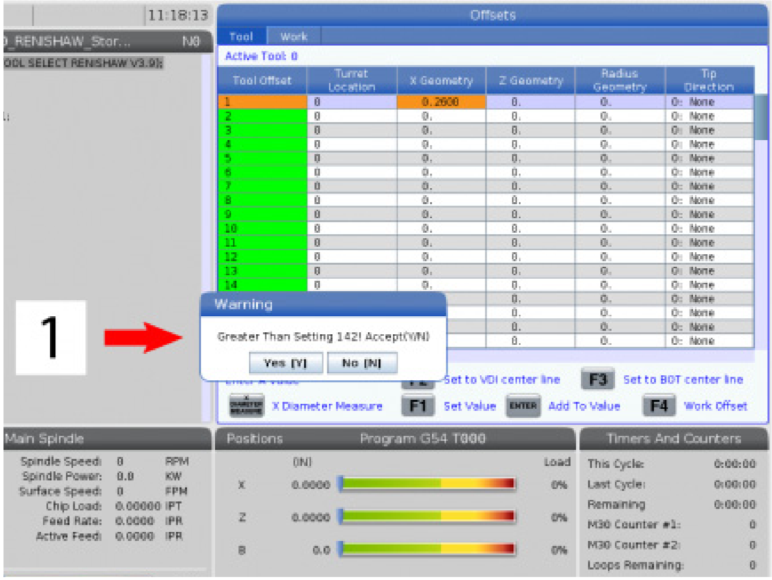
Make sure a turning-stick tool is installed in the first pocket.

Make sure the first pocket faces the spindle.

Jog the X and Z Axes to the center of the probe stylus [2].

Make sure that you have space [1] between the probe stylus [2] and the turning-stick tool.

4.



Push **[OFFSET]** one or two times to go to the TOOL GEOMETRY display.

Select the OFFSET 1 value.

Push 0. Push **[F2]**.

This removes the OFFSET 1 value.

If you get a warning message [1], Push**[Y]** to select YES.

Push**[.001]**.

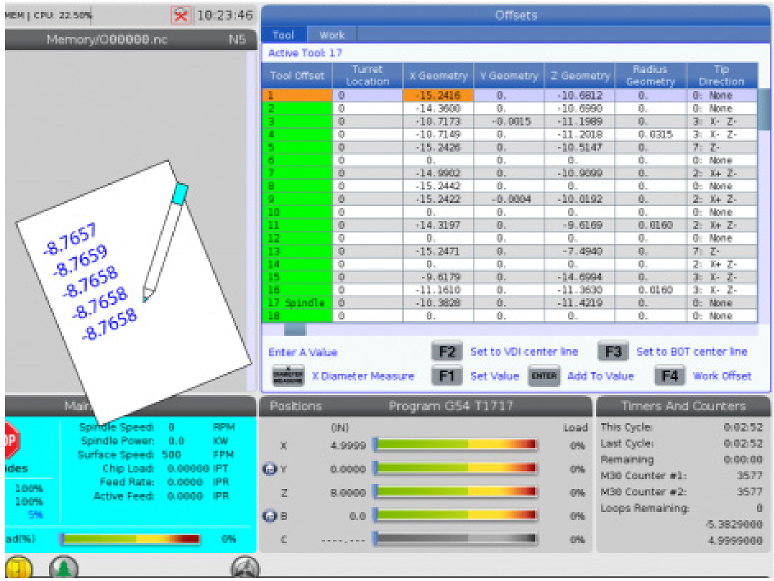
Push and hold the**[-X]** until the stick tool touches the probe.

NOTE: *You hear a beep sound when the stick tool touches the tool probe.*

Record the value in the OFFSET 1.

Jog the X Axis away from the ATP arm.Do steps 2, 3, and 4 four times.

5.



Compare the highest and lowest recorded values.

If the difference is more than 0.002 (0.05 mm), you must measure and adjust the 3/8-24" x 2" setscrew installed into the ATP arm.

The 3/8-24" x 2" setscrew is possible not tightened correctly. If this occurs, do the Automatic Tool Presetter (ATP) - Alignment sub-procedure.

Put the recorded values from step 1 into the OFFSET values for TOOL 1.

Use the M104 and M105 commands in MDI mode to make sure the ATP operates correctly.

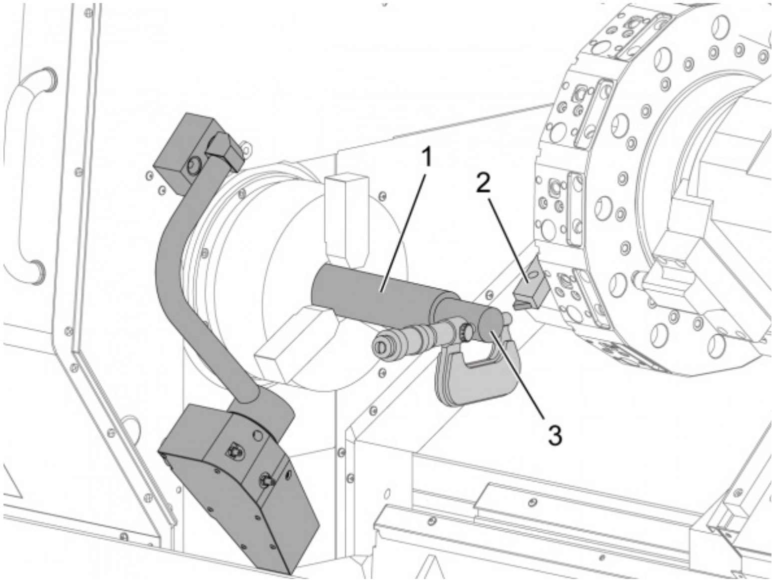
```
M104; (Tool Presetter Down)
```

```
M105; (Tool Presetter Up)
```

6.2.3 Automatic Tool Presetter (ATP) - Calibration

This procedure will show you how to calibrate the Automatic Tool Presetter.

1.



Install an OD turning tool in the turret tool 1 station [2].

Install a workpiece in the chuck [1].

Make a cut along the diameter of the workpiece in the negative Z-axis direction.

Push **[HAND JOG]**. Push**[.001]**. Hold down **[+Z]** to move the tool away from the part.

Stop the spindle.

Measure the diameter of the cut made on the workpiece [3].

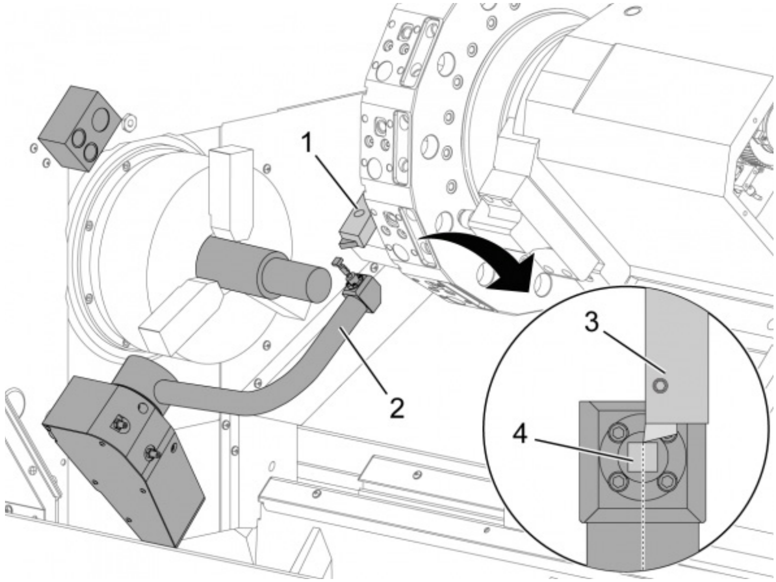
Push **[X DIAMETER MEASURE]** to put the value into the **[OFFSET]** column for the X-axis.

Enter the workpiece diameter.

Push **[ENTER]**. This adds the value to the **[OFFSET]** column value.

Record this value as a positive number. This is Offset A. Change Settings 59 through 61, 333, and 334 to 0.

2.



Move the tool away [1] to a safe position out of the ATP arm path [2].

Operate this code in MDI mode: M104.

This moves the ATP arm to the down position.

Jog the Z-axis to align the tool tip [3] with the center of the stylus [4].

Jog the X-axis to move the tool tip to 0.25" (6.4 mm) above the probe stylus.

Push**[.001]**.

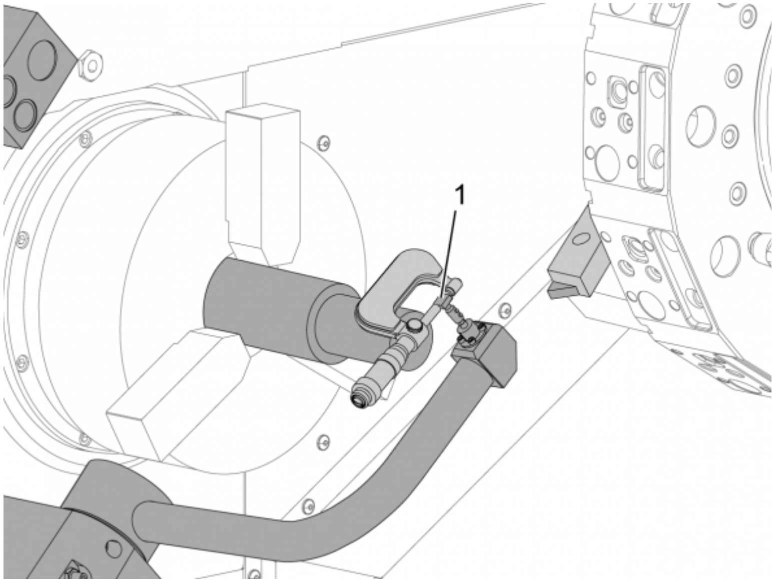
Hold down **[-X]** until the probe makes a "beep" sound and stops the tool.

Record the X-axis **[OFFSET]** column value as a positive number.

This is Offset B.Subtract Offset B from Offset A.

Enter the result as a positive value into Setting 59

3.



Measure the width of the stylus [1].

Enter this value as a positive number for Settings 63 and 334.

If the probe stylus is calibrated correctly, the values from **[X DIAMETER MEASURE]** and the value from the stylus are equal.

Multiply the probe stylus width by two.

Subtract that value from Setting 59.

Enter this value as a positive number into Setting 60.

Setting 333 will remain zero.

AFTER COMPLETING THIS TASK:

Change the below macro values to match the setting values.

- Setting 59 = #10582
- Setting 60 = #10583
- Setting 63 = #10585
- Setting 333 = #10584
- Setting 334 = #10585